

Worksheet — 1.5 Legal, Moral, Cultural & Ethical Issues — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Answer key

Activity 1

1 = C · 2 = D · 3 = B · 4 = A

Activity 2 — correct Act for each scenario

1. **Data Protection Act 2018 / GDPR** — breaches *storage limitation* (keeping data longer than needed).
2. **Computer Misuse Act 1990 — Offence 1** (unauthorised access; nothing was changed and there was no further intent).
3. **Copyright, Designs & Patents Act 1988** — distributing a copyrighted work without a licence (piracy).
4. **Computer Misuse Act 1990 — Offence 2** (unauthorised access *with intent* to commit a further offence, i.e. theft). *Accept a note that the DPA may also apply to the bank's data handling.*
5. **Regulation of Investigatory Powers Act 2000 (RIPA)** — interception by a public body without proper authorisation.
6. **Computer Misuse Act 1990 — Offence 3** (unauthorised modification — encrypting/altering files).

Activity 5 — model points

1. **Personal data:** information relating to an identifiable living individual (e.g. name, address, ID number, location, online identifiers).
2. **Data subject:** the living individual whom the personal data is about.
3. **Copyright** protects the *expression* of original works automatically (no registration), giving the rights-holder control over copying/distributing/adapting. A **patent** protects an *invention*, is **not** automatic — it must be applied for and granted.
4. **Digital divide:** the gap between those who have access to technology, the internet and the skills to use it, and those who do not (often poorer, rural, elderly or disabled people).
5. **GDPR** is the EU regulation; the **DPA 2018** is the UK law that implements GDPR in domestic law.

Activity 4 — model essay plan

FOR (does more harm than good) - *No empathy/human judgement* → unusual but valid cases get unfairly rejected → harms **individuals/applicants** → but a human can review borderline cases (appeal route). - *Algorithmic bias* → systems trained on biased data reproduce/amplify discrimination → harms **disadvantaged groups & society** → but bias can be audited and corrected, unlike hidden human prejudice. - *"Black box" / unclear accountability* → hard to explain or appeal a decision; unclear who is liable → harms **individuals & regulators** → but explainable-AI and logging can improve transparency.

AGAINST (does more good than harm) - *Speed and scale* → decisions made instantly on huge volumes of applications → benefits **businesses & consumers** (faster service) → but speed is worthless if the decisions are unfair. - *Consistency / no fatigue* → same rules applied to everyone, no tiredness or mood → benefits **society** (fairer in theory) → but only as fair as the data and rules behind it. - *Better outcomes in some domains* → e.g. medical diagnosis can spot patterns humans miss → benefits **patients** → but should support, not replace, the clinician.

Stakeholders: individuals/applicants · businesses/employers · disadvantaged groups · government/regulators · patients.

Justified conclusion (takes a side, with a condition): Automated decision-making does more good than harm **only if** it is used to *support* rather than fully replace human judgement, the training data is checked for bias, decisions are explainable, and there is always a route to human appeal. Without these safeguards, the harms (bias, lack of empathy, unaccountability) outweigh the benefits. A top-band answer reaches a clear, reasoned position rather than just stopping.

Activity 6 — model answers (balanced argument builder)

Stakeholder	Benefit (+)	Drawback (–)
Customers / shoppers	Shorter queues and faster checkout because many self-checkouts replace fewer staffed tills; potentially lower prices as the shop's wage bill falls.	Loss of privacy — their faces are scanned and stored; risk of being wrongly flagged as a shoplifter by a misidentification (facial recognition is less accurate for some ethnic groups); elderly/disabled shoppers may struggle without staff help.
Checkout staff	A minority may be retrained into higher-skilled roles (maintaining/supervising the machines), which can mean better pay.	Large-scale job losses/deskilling ; remaining roles may be fewer, insecure, and require retraining that not all staff can access.
The supermarket (business)	Lower long-term staffing costs; theft may fall; data on shopping patterns can improve stock control.	High up-front cost of the technology; reputational damage and possible legal action (DPA 2018 — biometric data is <i>special category</i> data) if the system misidentifies innocent customers or is breached.
Government / wider society	Possible productivity gains; a test case that informs regulation of biometrics in public spaces.	Normalises mass surveillance ; unemployment costs (benefits, retraining) shift to the state; widens inequality if displaced low-skill workers cannot find comparable work.

(a) Strongest answers usually argue **checkout staff** (livelihoods lost entirely, and the harm is concentrated on a group least able to absorb it) or **wrongly-identified customers** (accusation of theft is a serious individual harm). Credit any choice with a clear justification. (b) Safeguards (any one): human review before anyone is stopped/accused; keeping some staffed tills; consent/clear signage and short retention periods for facial data; independent accuracy audits across demographic groups; a retraining fund for displaced staff.

Activity 7 — precision answers

#	Act (full name + year)	CMA offence
1	Computer Misuse Act 1990	Offence 1 — unauthorised access; no modification, no further intent.
2	Computer Misuse Act 1990	Offence 3 — unauthorised modification / impairing operation of a computer (deleting files).
3	No Act broken	n/a — the access is authorised (signed contract), so the CMA is not engaged.
4	Computer Misuse Act 1990	Offence 2 — unauthorised access <i>with intent</i> to commit a further offence (theft/fraud with the card details).
5	Copyright, Designs and Patents Act 1988	n/a — using a copyrighted image without a licence.
6	Data Protection Act 2018 (GDPR)	n/a — processing personal data without consent/lawful basis; breaches fairness and transparency.
7	Regulation of Investigatory Powers Act 2000 (RIPA)	n/a — RIPA is the Act that <i>authorises</i> lawful interception by public bodies when the proper warrant/authorisation is obtained.

No offence: **scenario 3** — the single word is "**authorised**" (accept "permission/consent"). The CMA only criminalises *unauthorised* acts.

Challenge box

(a) Computer Misuse Act 1990 (the hacker gained unauthorised access to the servers) **and the Data Protection Act 2018 / GDPR** (the retailer failed in its duty of *integrity & confidentiality/security* by storing data unencrypted, and breached *storage limitation* by keeping it too long). *Two laws can apply to one event.* **(b)** The hacker has committed **Offence 1** (unauthorised access) and **Offence 2** (access *with intent* to commit a further offence — stealing/using the card details). *Accept that copying the data could also engage the DPA against whoever misuses it.*

Section B — model answers

Q1 [2] (AO1) — Any two DPA 2018/GDPR principles, 1 mark each, e.g.: - Data must be processed **lawfully, fairly and transparently**. - Data must be collected for a **specified, explicit purpose** and not reused incompatibly (purpose limitation). - Data must be **adequate, relevant and limited** to what is necessary (data minimisation). - Data must be **accurate** and kept up to date. - Data must not be kept **longer than necessary** (storage limitation). - Data must be kept **secure** (integrity and confidentiality). (*Do not accept vague answers such as "keep data safe from hackers" alone without the idea of a principle; accept sensible paraphrases.*)

Q2 [2] (AO2) - Computer Misuse Act 1990 (1 mark — full name required; "CMA 1990" acceptable). - **Offence 1 / Section 1 — unauthorised access** to computer material (1 mark). Since the hacker only *viewed* records with no modification and no evidence of intent to commit a further offence, offences 2 and 3 do not apply.

Q3 [3] (AO2) — Three linked points, e.g.: - The source code is an original work whose **copyright** belongs to FitPulse (code written in employment belongs to the **employer**, not the employee) (1). - The relevant legislation is the **Copyright, Designs and Patents Act 1988** (1). - Copying/reusing the code in a rival app

without a **licence/permission** would infringe FitPulse's exclusive rights to copy and adapt the work, so FitPulse could sue / seek an injunction (1).

Q4 [4] (AO2) — Two concerns, each identified (1) and developed with a consequence applied to FitPulse users (1), e.g.: - **Re-identification**: "anonymised" location data can often be de-anonymised (1), because a person's home and workplace uniquely identify them — so advertisers or anyone who buys the data could track *individual* FitPulse users' movements (1). - **Consent / purpose limitation**: users gave their data to track fitness, not for advertising (1); selling it for a new purpose without clear, freely given consent breaches transparency and purpose-limitation expectations under the DPA 2018 (1). - Also creditworthy: **security risk** — the more organisations hold the data, the greater the chance of a breach exposing users' daily routines (burglary/stalking risk); **profiling** — location reveals sensitive facts (place of worship, clinics attended).

Q5 [3] (AO2) — Any three, e.g.: - RIPA regulates how **public bodies** (police, security services) may carry out surveillance and obtain communications/communications data (1). - The police must obtain the **proper authorisation** (e.g. a warrant / sign-off at the required level) before FitPulse can lawfully be compelled to hand over the data; the request must be for a legitimate purpose such as preventing or detecting crime (1). - The Act can **compel organisations to hand over data** (and in some cases to disclose keys/decrypt protected information), and it makes interception **without** authorisation unlawful (1).

Q6 [4] (AO2) - Explanation (up to 2): the **digital divide** is the gap between those with access to devices/connectivity/skills and those without (1); here, customers without a capable smartphone are locked out of the *cheaper* insurance, so an existing disadvantage (age/income) is converted into a direct **financial penalty** — the divide widens (1). - Action (up to 2): any sensible mitigation identified (1) + how it reduces the effect (1), e.g. offer a **non-app route** (posted reports / SMS / in-branch sync) so no smartphone is needed; subsidise or loan compatible devices; guarantee the standard premium is not *raised* for non-participants; run digital-skills support for elderly customers.

Q7* [12] (AO3) — levels of response

Level	Marks	Descriptor (OCR style)
Level 3	9–12	A thorough discussion showing detailed knowledge of the ethical, legal and social issues. The answer is balanced , considering both good and harm across a range of stakeholders , and is consistently applied to FitPulse's scheme rather than generic. There is a well-developed line of reasoning which is clear and logically structured ; evaluative comments are substantiated (supported by consequences/evidence), and the answer reaches a justified conclusion .
Level 2	5–8	A reasonable discussion showing sound knowledge. Both sides are considered but the treatment may be uneven or only partly applied to the scenario; some points remain undeveloped or generic. There is a line of reasoning with some structure ; evaluative comments are present but only partially substantiated ; any conclusion may be weakly supported.
Level 1	1–4	A basic answer: a few relevant points, largely one-sided or generic , with little or no application to FitPulse. Little or no evaluation; the information is poorly structured.
—	0	No response or no response worthy of credit.

Indicative content — *good/benefits*: lower premiums reward and incentivise healthy behaviour (users, public health); insurers price risk more accurately (business); aggregate data could support medical research (society). *Harm/drawbacks*: those who refuse effectively pay a **penalty**, so "consent" is not freely given (users); discrimination against people whose data looks "unhealthy" through disability, poverty or shift work (disadvantaged groups); the digital divide excludes those without devices (Q6 link); continuous heart-rate/GPS data is intimate — breaches, function creep and resale risks (users/regulators, DPA 2018);

normalises surveillance-based pricing (society). *Stakeholders*: users who join, users who refuse or can't join, FitPulse, the insurer, regulators/government, wider society.

Model top-band answer (full prose):

FitPulse's scheme offers genuine benefits. Users who opt in receive lower premiums, and because the wristband gives the insurer accurate, continuous evidence of activity, pricing reflects real behaviour rather than crude averages — a healthy 60-year-old is no longer charged as if they were an average one. This can also nudge people towards exercise, since better data directly means cheaper insurance, which benefits public health and, in the long run, reduces claims — good for the insurer and arguably for society. For FitPulse, the partnership creates revenue that can fund better products.

However, the harms fall on those least able to avoid them. Because refusers pay the standard, higher price, the "choice" to share intimate health data is not freely given: economically, opting out is a penalty, which undermines the DPA 2018's requirement that consent be freely given and unambiguous. The scheme also discriminates structurally. A night-shift worker's disturbed sleep, or a disabled user's low step count, will look "unhealthy" regardless of anything they can control, so the algorithm converts existing disadvantage into higher premiums. As established in Q6, elderly and low-income customers without smartphones are excluded from the discount altogether, widening the digital divide. There are also serious data-protection risks: continuous heart-rate and GPS data reveals pregnancies, illnesses, places of worship and daily routines; a breach at either FitPulse or the insurer would be far more damaging than a leaked email address, and once shared, function creep — reuse for marketing or claim disputes — is hard for users to police.

Weighing these, the benefits are real but mostly accrue to the insurer, FitPulse and already-healthy, already-connected users, while the harms concentrate on vulnerable groups and are partly irreversible (leaked health data cannot be recalled). I therefore conclude the scheme currently does **more harm than good**, and would only become acceptable with safeguards: the non-sharing price must be the genuine baseline (a discount, not a penalty), data must be minimised and deleted on opt-out, the pricing algorithm must be audited for bias against disability and shift patterns, and a non-smartphone route must exist. With those conditions the balance could tip towards good; without them it does not.

(Section B total: 2 + 2 + 3 + 4 + 3 + 4 + 12 = 30 marks.)